

Construction Sites and Parking Tickets in Toronto*

[Write: a subtitle that states the main finding]

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[Write: three or four sentences. (1) What was done: 32 million Toronto parking tickets from 2010 to 2025 were placed on street blocks, and blocks where construction started were compared with matched nearby blocks without construction. (2) What was found: construction blocks received about 0.4 more tickets per block per month, roughly 8 per cent more, and the gap grew over the first two years. (3) Why it matters.]

1 Introduction

Write: paragraph on the broader context. Why parking enforcement matters in Toronto: the number of tickets each year (about 2,070,000 a year since 2022), the money involved, curb space as a scarce resource (Shoup 2005), and fines as a way to change behaviour (Becker 1968).

Write: paragraph on the gap and what was done. Construction sites take curb space and bring trucks, but whether they change where tickets are written is not known. This paper places 32,211,737 tickets from 2010 to 2025 on the street network and compares 5,484 blocks where construction started with matched nearby blocks that had none, using a difference-in-differences design (Card and Krueger 1994; Angrist and Pischke 2009).

Write: paragraph on what was found and why it matters. Construction blocks received 0.44 more tickets per block per month than their matched blocks (95% confidence interval 0.29 to 0.59), about 8% more, and the gap grew the longer construction went on. Be

*Code and data are available at: <https://github.com/FurkanDanisman/toronto-parking-tickets>.

specific; mention the caveat from Section 4.

The remainder of this paper is structured as follows. Section 2 describes the parking ticket and building permit data, how they are measured, and where tickets are written across the city and the day. Section 3 explains how blocks with construction are compared with similar blocks without it and presents the results. Section 4 discusses what the results mean and their limitations. Section A in the appendix describes how the data were cleaned and matched to the street network, and Section B reports alternative versions of the comparison.

2 Data

2.1 Source and context

Write: the data come from Open Data Toronto, downloaded with the `opendatatoronto` package (Gelfand 2025). Parking tickets (City of Toronto 2026f); building permits (City of Toronto 2026b, 2026c); address points (City of Toronto 2026a); street centreline (City of Toronto 2026g); former municipality boundaries (City of Toronto 2026d); green spaces (City of Toronto 2026e). Who publishes them and why; privacy (licence plates are removed and ticket numbers masked); why 2020 and 2021 are excluded (the pandemic changed parking and enforcement); similar datasets that could have been used and why they were not (for example, road restrictions exist only as a live feed, and utility cut permits only cover 2024 onward).

The analysis uses R (R Core Team 2024) with the `tidyverse` (Wickham et al. 2019), `arrow` (Richardson et al. 2024), `sf` (Pebesma and Bivand 2023), `jsonlite` (Ooms 2014), `fixest` (Bergé 2018), `testthat` (Wickham 2011), `here` (Müller 2020), and `knitr` (Xie 2024). Graphs are made with `ggplot2` (Wickham 2016) and `hexbin` (Carr et al. 2024), and tables with `tinytable` (Arel-Bundock 2024).

2.2 Measurement

Write: how an event in the world becomes a row. A parking enforcement officer (or police officer) sees a vehicle parked against a rule, issues a ticket with the date, time, infraction, fine and a location written as an address or an intersection; the city publishes the ticket with the plate removed. What this misses: violations nobody ticketed, so the counts measure enforcement as much as behaviour; tickets later cancelled or disputed are still counted. For construction: a permit is issued, but its “completed” date is administrative, so active periods are approximated (see Section A).

Table 1: Each parking ticket records when and where it was issued, the rule that was broken, and the fine. Ticket numbers are partly masked and licence plates are not published.

Variable	Description	Example
Ticket number	Ticket identifier with the first digits masked	***65822
Date	Day the ticket was issued	1 January 2025
Time	Time on a 24-hour clock	00:02
Infraction code	Number identifying the rule that was broken	29
Infraction description	Short text version of the rule	Park prohibited time no permit
Fine	Set fine in dollars	\$45
Location	Street address, or a street and a cross street	Near 77 Portland St
Province	Province or state of the licence plate	ON

2.3 Parking tickets

Table 1 lists what each ticket records.

Write: describe Table 2: about two million tickets a year, the share placed on a street block, how the average fine changed; note that the published 2013 file stops on 18 November 2013.

Write: describe Table 3: the most common rules broken in 2025, their fines, and which happen overnight.

Write: describe Figure 1: where tickets are written at different times of day. Overnight tickets spread across the whole city; during working hours they concentrate downtown and along Yonge Street.

Table 2: Toronto issues about two million parking tickets a year, and more than 85 per cent can be placed on a specific street block. The average fine was stable through the 2010s and rose sharply in 2024 and 2025. The 2013 file ends on 18 November 2013.

Year	Tickets	Placed on a street block	Average fine	Total fines (\$ millions)
2010	2,721,462	86%	\$40	110.0
2011	2,805,492	86%	\$41	114.2
2012	2,746,154	86%	\$41	111.5
2013	2,351,358	87%	\$41	95.8
2014	2,484,983	87%	\$44	109.0
2015	2,168,493	86%	\$46	99.9
2016	2,254,761	87%	\$49	109.7
2017	2,144,226	87%	\$50	106.2
2018	2,044,288	87%	\$49	100.2
2019	2,213,939	86%	\$49	108.4
2022	1,821,887	86%	\$53	96.3
2023	2,256,360	88%	\$54	122.4
2024	2,147,159	88%	\$68	146.0
2025	2,051,175	88%	\$80	164.7

Table 3: Eight rules account for most tickets. Parking on private property is the most common; tickets for parking without a permit and for staying longer than three hours are mostly written overnight, when permit parking rules apply.

Code	Rule	Fine	Tickets	Share of all tickets	Written overnight
3	Park on private property	\$75	387,901	19%	32%
5	Park-signed hwy-prohibit dy/tm	\$65	286,967	14%	8%
29	Park prohibited time no permit	\$45	273,337	13%	45%
207	Park machine-reqd fee not paid	\$50	270,929	13%	0%
9	Stop-signed hwy-prohibit tm/dy	\$120	144,105	7%	4%
2	Park - longer than 3 hours	\$40	139,059	7%	63%
403	Stop-signed highway-rush hour	\$190	75,710	4%	0%
312	Parking mach-not used/no fee	\$75	63,705	3%	0%

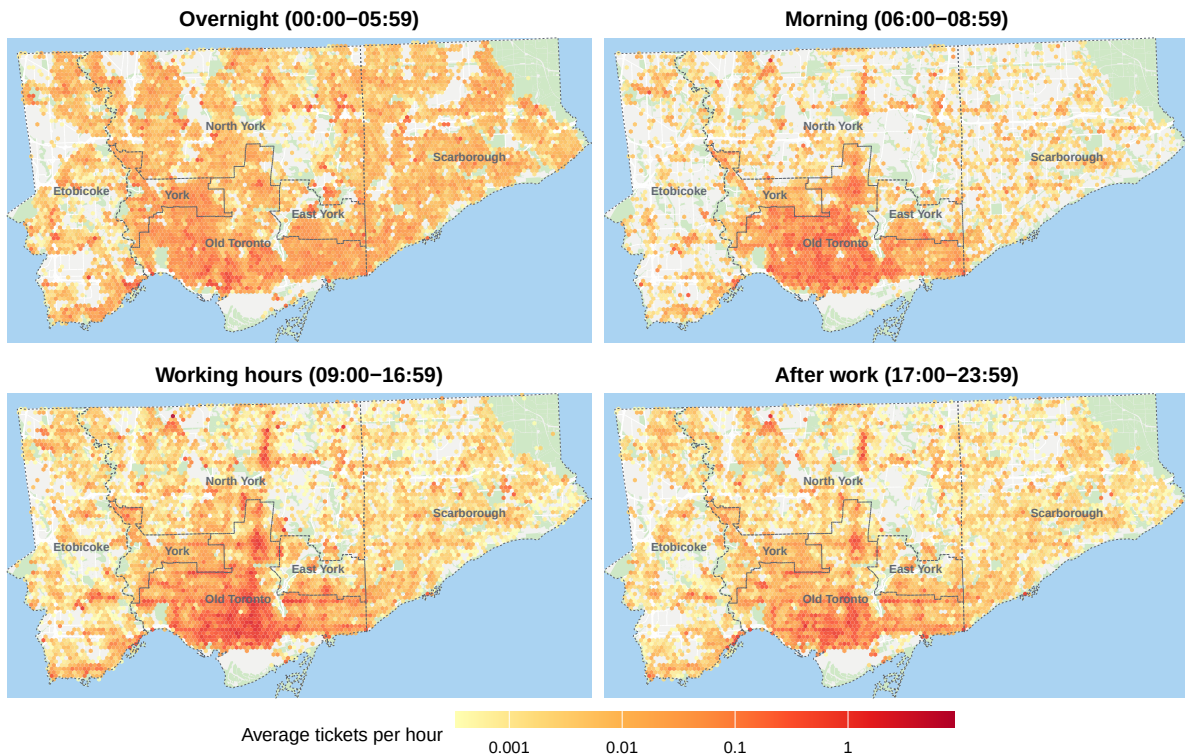


Figure 1: Overnight tickets are spread across the city, while working-hours tickets concentrate downtown. Each hexagon is 300 metres across; its colour is the average number of tickets issued there per hour in that time window on a 2025 day. Dashed lines are the six former municipalities amalgamated in 1998. The map is rotated so Toronto's street grid runs horizontally.

Table 4: Building permits identify more than 30,000 new buildings and demolitions active between 2010 and 2025, and about nine in ten can be placed on a street block. Demolitions are short; new buildings are counted as active for up to three years.

Type of work	Sites	Placed on a street block	Typical months active
Demolition	12,628	92%	6
New Building	18,716	91%	36

2.4 Construction sites

Write: building permits identify construction sites; only new buildings and demolitions are kept, because interior, plumbing and mechanical work does not affect the street. Describe Table 4.

3 Construction and parking tickets

3.1 Comparing blocks with and without construction

Write: explain difference-in-differences in plain language. A block where construction starts (the treatment group) is compared with the average of five nearby blocks with no construction (the control group). Controls are within 1.5 km but more than 250 m away (so construction does not spill onto them), on the same kind of street, with no construction from 12 months before to 24 months after the start, and with the most similar number and trend of tickets in the year before (Stuart 2010). Each construction block and its controls form a “stack” observed over the same calendar months (Wing et al. 2024). The effect is how much the gap between the two groups changes after construction starts, compared with the year before. Explain why the year before matters (the groups should move in parallel before construction) and the risk of regression to the mean in matched designs (Daw and Hatfield 2018; Roth et al. 2023).

The comparison is estimated with the regression

$$y_{it} = \alpha_i + \gamma_{s(i),t} + \beta D_{it} + \varepsilon_{it},$$

Write: explain each symbol in words. y_{it} is the number of tickets on block i in month t ; α_i is each block’s usual number of tickets; $\gamma_{s(i),t}$ is anything a construction block shares with its own control blocks in that calendar month, such as a citywide enforcement push

or a snowy February; D_{it} is 1 for a construction block after its construction started and 0 otherwise; β is the effect, measured in extra tickets per block per month; ε_{it} is everything else. Explain the confidence interval and p-value in plain words.

3.2 Results

Write: describe Figure 2: the two groups move in parallel before construction; afterwards the control group continues a slow decline shared by the whole city while construction blocks rise.

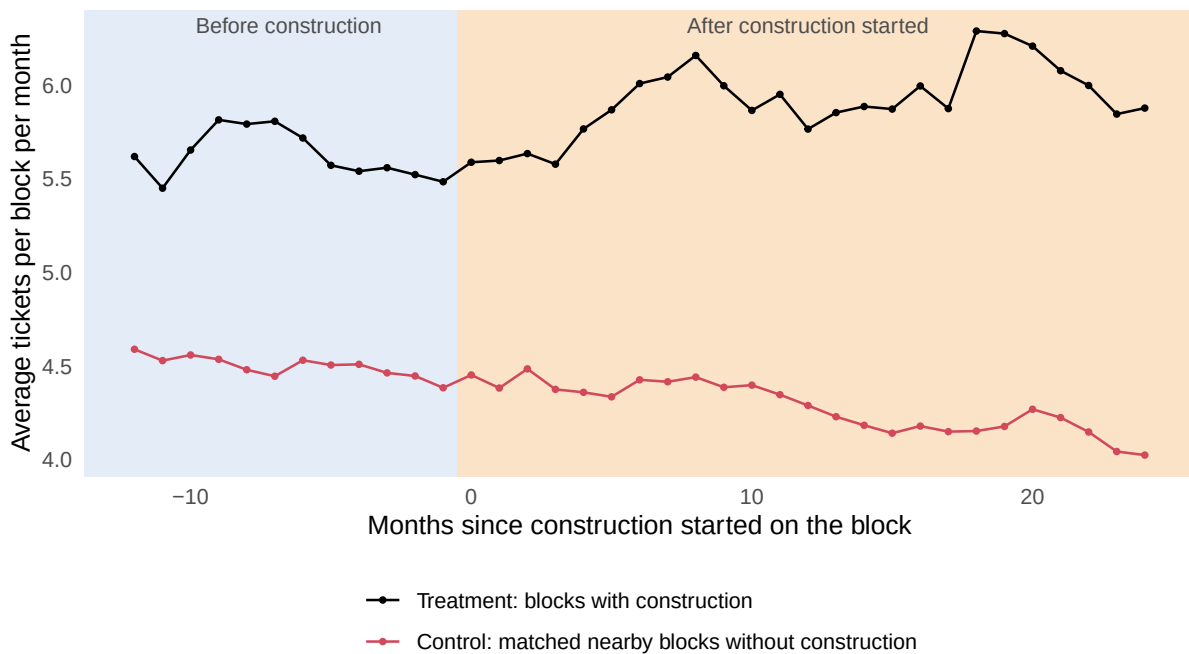


Figure 2: Tickets on blocks where construction started rose, while similar nearby blocks without construction continued a slow decline. Each line is the average number of tickets per block per month; the control line averages each construction block's five matched nearby blocks, so every construction block and its controls count once.

Write: describe Figure 3 and the overall result: 0.44 extra tickets per block per month (95% confidence interval 0.29 to 0.59), about 8% of the tickets a construction block received before; the gap grows from the first six months to the second year; the year before is not significantly different.

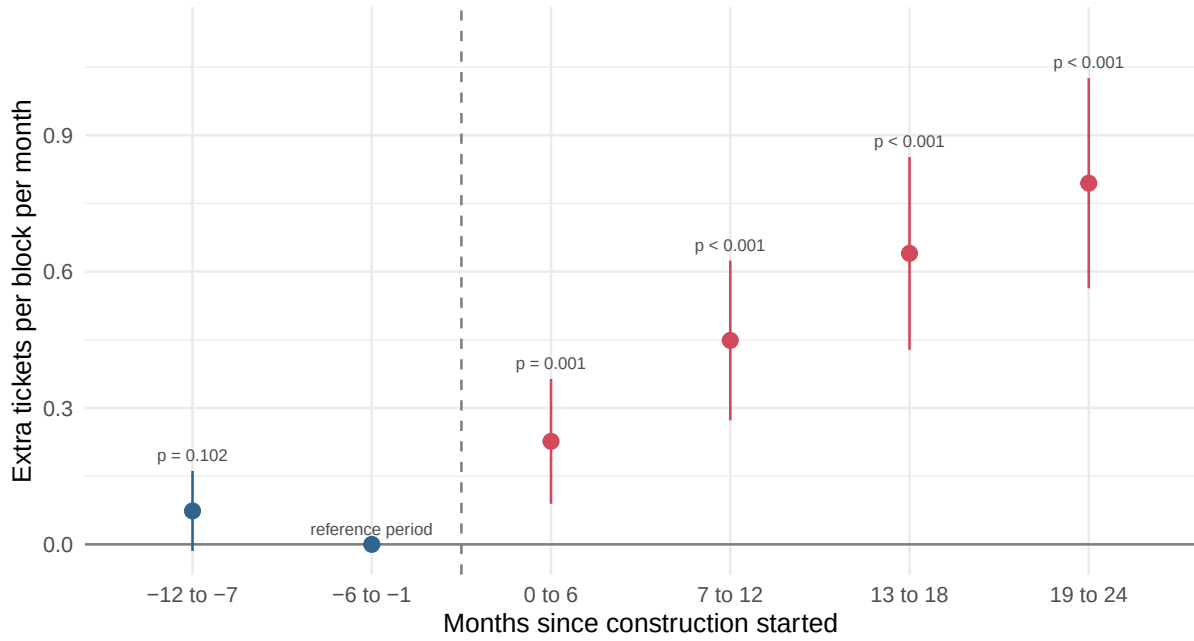


Figure 3: The gap between construction blocks and their matched nearby blocks grew steadily after construction started. Each point is the change in the gap, in tickets per block per month, relative to the six months before construction; vertical lines are 95 per cent confidence intervals. Before construction the gap was not significantly different from the reference period.

4 Discussion

Write: what we learn. Possible explanations: construction takes curb space and brings trucks and workers who park where they should not; enforcement may follow complaints.

Write: limitations. When the control group is nearby blocks that get construction later, the effect is smaller and not statistically significant (Table 5), so part of the gap may reflect the kinds of places where construction happens. Tickets measure enforcement as well as behaviour. Construction dates are approximate. About 13% of tickets could not be placed on a block.

Write: next steps.

Appendix

A Data cleaning

Write: from these facts:

- Tickets were read directly from the city’s ZIP files, because reading them through `opendatatoronto` silently returned only 1.42 million of the 2.75 million tickets for 2012. The 2010 file is stored in a different text encoding (UTF-16) and was converted.
- The published 2013 file ends on 18 November 2013; November and December 2013 are left out of the comparison.
- Each ticket’s street address was standardised (for example “AVENUE” to “AVE”) and matched to the city’s address points, then to the nearest street block; tickets written at intersections have no street number and could not be placed.
- About 3 per cent of building permits record a completion date before their issue date and were dropped. Because completion dates are administrative, a site counts as active for at most six months for a demolition and three years for a new building.

B Other versions of the comparison

Table 5 reports the overall result under reasonable alternative choices.

Write: explain the Poisson row: it measures the average percentage change per block, which gives quiet blocks with a few tickets a year as much weight as busy blocks, and finds no change.

Table 5: Construction blocks received more tickets than matched nearby blocks in most versions of the comparison, but the gap is smaller and not statistically significant when the control blocks are ones that get construction later. Estimates are extra tickets per block per month over the 24 months after construction started.

Comparison	Blocks	Effect	95% interval	Per cent of before	p-value
Nearby blocks, no construction in window	5,484	0.44	0.29 to 0.59	7.8%	< 0.001
Nearby blocks that get construction later	4,522	0.16	-0.08 to 0.39	2.7%	0.183
Only blocks observed in every month of the window	2,871	0.42	0.25 to 0.60	8.0%	< 0.001
Percentage change per block (Poisson)	5,484			-0.7%	0.577

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